

RIP-OSPF:

=====

Redistribute RIP Routes in to OSPF:

r2(config)#router ospf 5

r2(config-router)#redistribute rip metric 50000 subnet

OR

r2(config-router)#redistribute rip

r2(config-router)#exit

r2(config)#do write

Redistribute OSPF Routes into RIP:

r2(config)#router rip

r2(config-router)#redistribute ospf 5 metric 3

=====

Redistribute OSPF into EIGRP:

r2(config)#router eigrp 200

r2(config-router)#redistribute ospf 3 metric 1 1 1 1 1

Redistribute EIGRP into OSPF:

r2(config)#router ospf 3

r2(config-router)#redistribute eigrp 200 metric 50000 subnet

OR

r2(config-router)#redistribute eigrp 200

=====

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Access Control List:

Controlling the nw traffic passing through the router
it is like a firewallIncoming/Inbound Traffic
Outgoing/Outbound Traffic

Types of ACL

1. Standard
2. Extended

Ways to configure ACL

1. Numbered ACL

- | | | |
|-------------|---------|----------------|
| a. Standard | 1-99 | extended range |
| 1300-1999 | | |
| b. Extended | 100-199 | |
| 2000-2699 | | |

2. Named ACL

- a. Standard
- b. Extended

Rule: Permit Statements should be written after All Deny Statements.

access-list 140 deny host 192.168.1.2 host 192.168.1.3

access-list 140 deny host 192.168.1.4 host 192.168.1.5

access-list 140 permit ip any any

=====

Numbered ACL---Standard

Standard ACL:

1. preferably configured on Destination router.

2. can read / specify only source address.
3. can block entire network, cannot be used to block specific ip/port/service.

Extended ACL:

1. preferably configured on source router.
2. can read both source address & destination address
3. can be used to block specific ip, entire network, port/service.

Standard numbered ACL (creating rules):

```
=====
config# access-list 1 deny host 192.168.1.2          explicit deny
      access-list 1 permit any                      router will do
implicit deny for all the traffic
```

to apply a acl rule:

```
explicitly allow
config# int s0/0
      # ip access-group 1 IN
```

OR

to apply a acl rule:

```
config# int fa0/0
      # ip access-group 1 OUT
```

Extended Numbered ACL:

```
pc0(10.0.0.2/24)---not ping --pc3 (10.0.1.2/24)
```

To configure Extended ACL:

```
=====
config# access-list 120 deny ip host 10.0.0.2 host 10.0.1.2
config# access-list 120 permit ip any any
config# int fa0/0
      ip access-group 120 in
      do show access-list

      go to pc0-----cmd-----ping 10.0.1.2          (no pinging)
```

Extended ACL:

1. preferably configured on source router.
2. can read both source address & destination address
3. can be used to block specific ip, entire network, port/service.

LAB1: Standard ACL

```
pc0(192.168.1.2)----->pc1(192.168.1.34)
```

Middle Router:

```
=====
create ACL rule          config# access-list 5  deny host 192.168.1.2
#explicit deny

#implicit deny          config# access-list 5 permit any
#explicit allow

      to apply rule      #int fa0/0
                        # ip access-group 5  OUT
```

To remove ACL: # conf t

```
config# no access-list 5
# int fa0/0
```

```
=====
icmp ping
ip all services(complete block)
tcp 80
udp

#no ip access-group 5 out
```

LAB2: Extended ACL
pc0(192.168.1.2)---ping---->pc1(192.168.1.34)

Source Router(First):

```
192.168.1.34 config# access-list 110 deny icmp host 192.168.1.2 host
config# access-list 110 permit ip any any

config# int fa0/0
# ip access-group 110 in

or
config# int s0/0
# ip access-group 110 out
```

To remove ACL

```
config# no access-list 110
config# int fa0/0
# ip access-group 110 in
```

```
=====
LAB3: Extended ACL block laptop user from sales dept ---->not to access web
service--->http/https(80/443)-->tcp
Block pc
```

Source Router(Sales Users Router):

```
192.168.1.4 eq 80 config# access-list 130 deny tcp host 192.168.1.46 host
#access-list 130 permit ip any any

#int fa0/0
#ip access-group 130 in
#do sh access-list
```

```
=====
LAB4: Extended ACL block entire sales dept users(192.168.1.32/28)---->not to access web
service(192.168.1.4)--->http/https(80/443)-->tcp
Block pc
```

Source Router(Sales Users Router):

```
192.168.1.4 0.0.0.0 eq 80 config# access-list 130 deny tcp 192.168.1.32 0.0.0.15
#access-list 130 permit ip any any

#int fa0/0
#ip access-group 130 in
#do sh access-list
=====
```

192.168.1.1 host

access-list 30 deny host 192.168.1.1

or

access-list 30 deny 192.168.1.0 0.0.0.255
any

0--exact match, 255 ---

R1

192.168.1.2 D
192.168.2.1 A
192.168.1.4 D
192.168.5.2 A
192.168.1.1 D

172.16.0.1 A
172.16.0.255 A

172.16.0.1 0.0.0.0

access-list 15 deny 172.16.96.0 0.0.3.255

172.16.96.0 0-255
172.16.97.0
172.16.98.0
172.16.99.0

172.16.97.250 D
172.17.97.250 A
172.16.100.250 A
172.18.97.250 A

access-list 15 deny 172.16.128.0 0.0.15.255

172.16.128.0

172.16.143.0

access-list 15 deny 172.16.1.0 0.0.0.15

172.16.1.1 D
172.16.1.5 D
172.16.1.15 D
172.16.1.16 A
173.16.1.1

=====

LAB5: Extended ACL block entire Prod dept users(192.168.1.0/28)---->not to access FTP
service(192.168.1.66)---->tcp(20 & 21)

Source Router(Prod Users Router):

config# access-list 150 deny tcp 192.168.1.0 0.0.0.15
192.168.1.66 0.0.0.0 eq ftp

```
#access-list 150 permit ip any any
```

```
#int fa0/0
```

```
#ip access-group 150 in
```

```
#do sh access-list
```

```
=====
LAB6: Extended ACL      block all the services for prod users for network
(192.168.1.32/28) & block entire Prod dept
users(192.168.1.0/28)---->not to access FTP service(192.168.1.66).
```

```
192.168.1.66  0.0.0.0  config# access-list      150  deny  tcp   192.168.1.0   0.0.0.15
eq ftp
# access-list 150  deny  ip   192.168.1.0   0.0.0.15
192.168.1.32 0.0.0.15
```

```
# access-list 150 permit ip any any
```

```
#int fa0/0
```

```
#ip access-group 150 in
```

```
#do sh access-list
```

```
=====
LAB7: Extended ACL      Permit statement LAB
```

```
Web Server 192.168.1.4      permit      pc2      192.168.1.35
explicit allow
```

```
=====
implicit deny (all services)
```

2. Named ACL

a. Standard

b. Extended

```
access-list
```

Advantages of Named ACL

1. No need to remember ACL range values

2. Allow to remove any ACL statement, unlike numbered ACL.

3. Use of name while writing ACL rules helps to identify the purpose of rule easily.

```
access-list      extended  deny  sales   192.168.1.0   0.0.0.15   192.168.1.66   0.0.0.0
eq ftp
```

```
access-list      150  deny  tcp   192.168.196.0   0.0.0.15   192.168.1.66   0.0.0.0   eq
ftp
```

```
no access-list      150  deny  tcp   192.168.1.0   0.0.0.15   192.168.1.66   0.0.0.0
eq ftp
```

```
access-list      150  deny  tcp   192.168.1.0   0.0.0.15   192.168.1.66   0.0.0.0   eq ftp
```

```
access-list      150  deny  tcp   192.168.1.0   0.0.0.15   192.168.1.66   0.0.0.0   eq ftp
```

```
access-list      150  deny  tcp   192.168.1.0   0.0.0.15   192.168.1.66   0.0.0.0   eq ftp
```

```
access-list      150  deny  tcp   192.168.1.0   0.0.0.15   192.168.1.66   0.0.0.0   eq ftp
```

```
access-list      150  deny  tcp   192.168.1.0   0.0.0.15   192.168.1.66   0.0.0.0   eq ftp
```

```
access-list      150  deny  tcp   192.168.1.0   0.0.0.15   192.168.1.66   0.0.0.0   eq ftp
```

```
access-list      150  deny  tcp   192.168.1.0   0.0.0.15   192.168.1.66   0.0.0.0   eq ftp
```

```
access-list      150  deny  tcp   192.168.1.0   0.0.0.15   192.168.1.66   0.0.0.0   eq ftp
```

```
access-list      150  deny  tcp   192.168.1.0   0.0.0.15   192.168.1.66   0.0.0.0   eq ftp
```

```

access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp
access-list 150 deny tcp 192.168.1.0 0.0.0.15 192.168.1.66 0.0.0.0 eq ftp

```

```

=====
LAB1: Named Standard ACL----->destination
pc0(192.168.1.2)----->pc1(192.168.1.34) 35 46
Middle Router:
=====

```

```

config# ip access-group standard block-pc0
(config-std-nacl)# deny host 192.168.1.2
                # permit any
                exit
config# int fa0/0
# ip access-group block-pc0 out
#do show access-list

```

To remove:

```

config# ip access-group standard block-pc0
(config-std-nacl)# no deny host 192.168.1.2
                #no permit any
                exit
config# int fa0/0
# no ip access-group block-pc0 out
#do show access-list

```

```

=====
Named ACL--Extended:
=====

```

PC0 XX PC2 (192.168.1.2 should not ping with 192.168.1.34)

On Source Router:

```

(config)#ip access-list extended block-pc0
(config-ext-nacl)#deny ip host 192.168.1.2 host 192.168.1.34
(config-ext-nacl)#permit ip any any
(config-ext-nacl)#exit
(config)#int fa0/0
(config-if)#ip access-group block-pc0 in
(config-if)#end
#show access-lists

```

```

=====
NAT      Network Address Translation      service/protocol--- Translates Private IP
Add into Public IP Add and vice-versa

```

IANA---->IPs & ASN

	Private IP	Public IP
1.	local address	1. Global address
2.	unregistered add	2. registered address
3.	used in LAN(inside Organization)	3. used over internet

4. free of cost
5. cannot carry data over internet

4. buy ip from ISP
5. send data over internet

A 10.0.0.0--10.255.255.255

A 1-126

B 172.16.0.0-172.31.255.255

B 128-191

C 192.168.0.0-192.168.255.255

C 192-223

Types of NAT

- | | |
|---|--|
| 1.Static
ip) | one to one mapping (one pvt ip---one pub |
| 2.Dynamic | dynamic mapping of pvt ip with public ip |
| 3.PAT (NAT with Overload)
Port Address translation | Multiple pvt ip ---one public ip |

=====

2³²===== IPv4

Adv of NAT

1. Saves the available IP Address (IPv4)
2. Enhances security

Disadvantages:

1. NAT address translation leads to delay in data transmission.
2. All routers should support NAT who are used in Network.

- | | |
|----------------------------|--|
| 1. Static | one-to one mapping |
| 2. Dynamic | Pvt IPs are dynamically mapped with |
| available public add | |
| 3. PAT (NAT with Overload) | multiple pvt add----->mapped with one or |
| more public add | |

=====

Static NAT Configuration:

On ISP Router: static Routing

```
config# ip route 200.0.0.0 255.255.255.252 s0/0
```

ON NAT-Router: Default Routing

```
config# ip route 0.0.0.0 0.0.0.0 s0/0/1
```

ON NAT-Router:

```
config# int fa0/0
      ip nat inside          #to translate Public IP into Pvt IP
      exit
config# int s0/0/1
      ip nat outside         #to translate Pvt IP into Public IP
      exit
config# ip nat inside source static 192.168.1.66 200.0.0.1
config# ip nat inside source static 192.168.1.67 200.0.0.2
```

To verify NAT Table:

```
do show ip nat translations
```

To remove Static NAT entries:

```
=====
config# int gi0/0
      no ip nat inside
      exit
config# int s0/0/1
      no ip nat outside
      exit
config# no ip nat inside source static 192.168.1.66 200.0.0.1
```

```
config# no ip nat inside source static 192.168.1.67 200.0.0.2
```